Task: Work on your group presentations.
Focus on how you will present during the group presentations

At the end of 30 minutes
Present your draft presentation and outline
Decide on the following

- How many presenters (more marks for better group participation)
- What you want to present, which artefacts
- How you want to present it
- Present what you have already completed for initial feedback

Slot 2

PPT guidelines

Website (first) / Power Point Presentation

Wireframe (first) / Power Point Presentation

- Problem statement / Design idea
- User persona
- User journey map (2 route?)
- Design metaphor
- Contextual Ref and Presentation
- SUSS – rating (check)
- Future upgrade (?)

BMI5206 Group Project

- Create biomedical based application
- Build a prototype (no need complete coding)
- Eg; consumer health / patient health focus

Grp Name: The Design Thinkers

Team Lead: Matthew

Project Management Style: Agile

Project Management tool: Kanban / Miro

How often we meet: before/after physical lectures for f2f, online for the rest of the weeks

Conflict management style: Get together and discuss any issues so we can resolve them amicably

-vote if unable to come to agreement. Use poll option

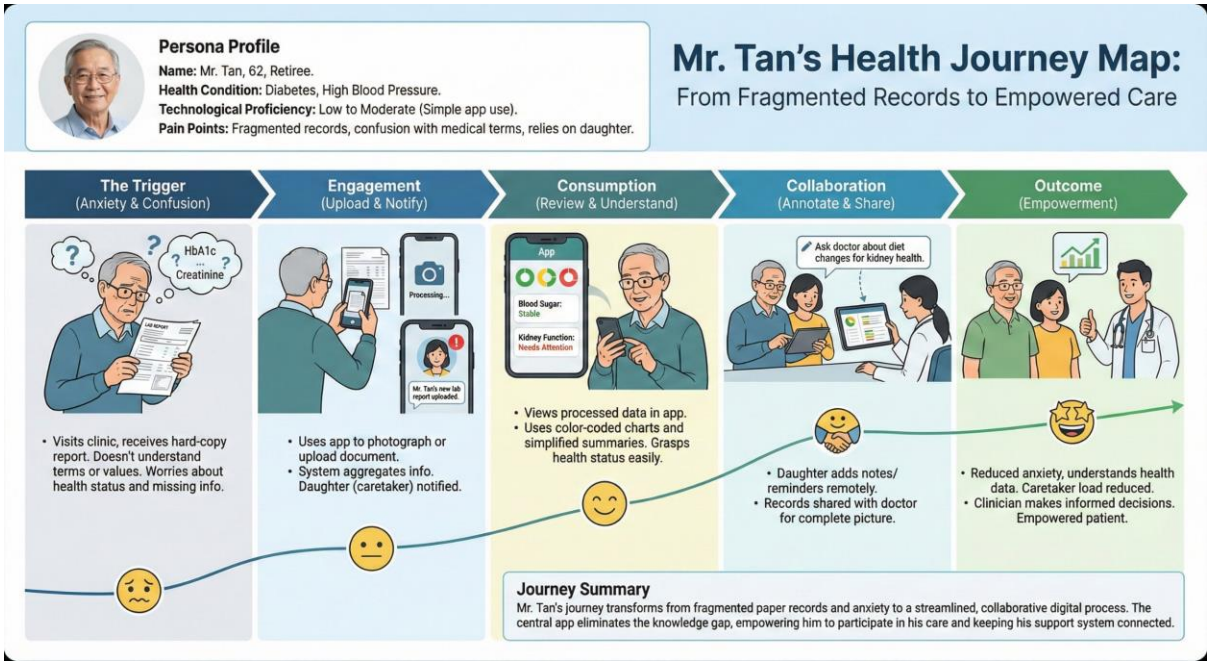
Problem idea: Fragmented patient EHR record

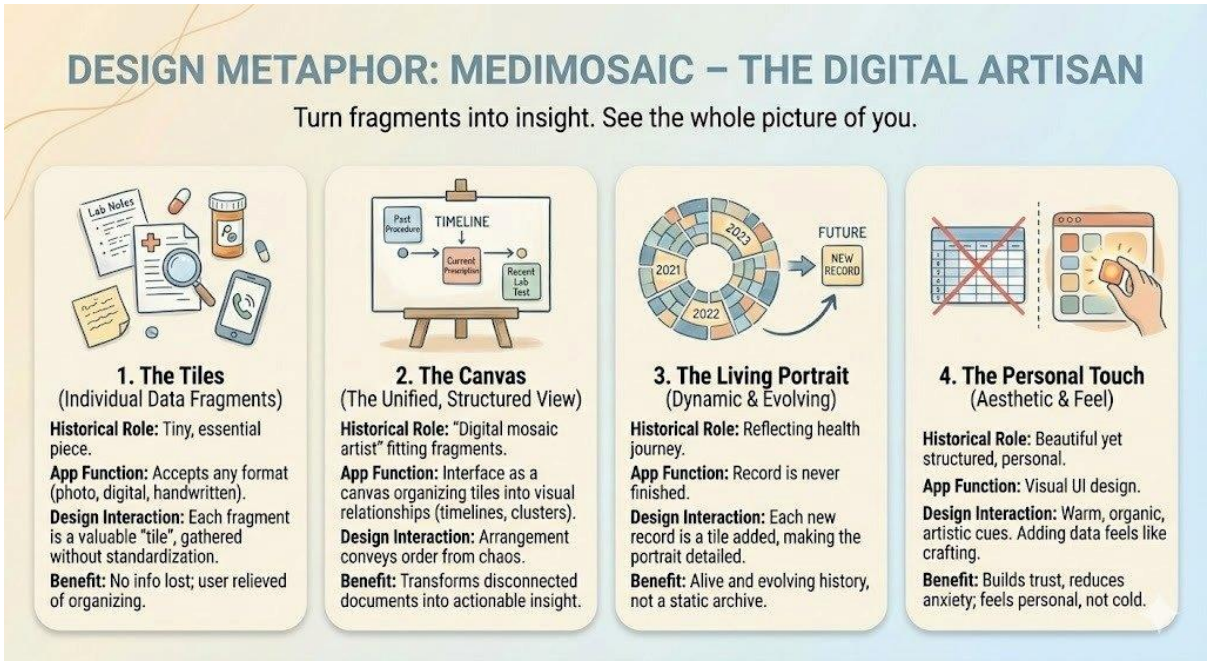
Project Idea: EMR system is not very user intuitive

Problem statement

Patient medical record data is frequently fragmented across multiple platforms, including different electronic health record (EHR) systems, handwritten notes from healthcare institutions without EHR systems, and patient-generated health data such as home blood pressure or blood glucose monitoring. This fragmentation makes it difficult for patients to understand their own health information, limits continuity of care across institutions, and increases cognitive burden for both patients, caretakers and clinicians. Patient education is often insufficient or poorly contextualised, further reducing the usefulness of available health data.

The proposed solution is a patient-centred mobile application that enables medical providers, patients and their caretakers to access and aggregate health information across medical institutions, upload pictures or PDF documents of health documents, and upload home health monitoring data. Patients will receive explanations and educational support tailored to their unique health journey.





People/Users

Patients and caretakers

Medical providers – doctors, nursing, allied health, pharmacy

Issues: ability to use IT/apps, Health literacy

Activities:

Uploading home monitoring and health documents

Understanding lab results and health issues

Learning about medical conditions and aftercare

Context

Patient care across the healthcare journey including before and after medical encounters

Management of chronic diseases

Disjointed healthcare systems

Technology

Mobile application with secure cloud-based storage that aggregates health records across institutions, supports manual and automated home monitoring data upload, visualises trends over time, and provides plain-language, context-aware explanations for patients

Human-centred principles such as minimal data entry burden, clear visual hierarchy, avoidance of alert fatigue, support for error recovery, and accessibility through appropriate font size, contrast, and language simplicity

Safe

Get Clarification and avoid misunderstanding

Centralizing fragmented records reduces errors from missing information, like incorrect medications, duplicate tests, or overlooked lab results

Timely

Eliminating knowledge gap

Patient has faster access to medical records

Effective

Benefits the patients and help them to get better understanding with their clinicians if needed

Compiles patient data into one repository so clinicians can make effective, evidence-based decisions with all necessary information on hand

Efficient

Being able to get all the information on one app, it is helpful to visualize and understand the issues as a whole

Consolidating fragmented medical records prevents the need for unnecessary testing or procedures, allowing patient to be treated more efficiently

Equitable

Applicable to multiple patients profile and background, support patients of varying digital literacy, language, and socioeconomic backgrounds

Patients moving between multiple institutions or regions have equitable access to their own health information

Patient-Centered

Can alter based on individual patient's needs and preference, decrease the mental burden on the patient and caregivers, user friendly

Tailored tips, reminders, and insights based on individual data and health goals.

Patients can better participate in consultations armed with full health info and trends.

Personas

Persona 1: Patient (Mr. Tan)

Age: 62

Occupation: Retiree

Tech savviness: Low to moderate

Mr. Tan is a 62 year old patient with common chronic diseases in older adults. He has diabetes and high blood pressure. He visits different hospitals and clinics. His medical records are spread across many systems and paper reports. He can take photos with his phone and use simple app functions, but he cannot manage complex digital tools. He does not understand medical terms or lab numbers well. When he receives test results, he often feels confused and anxious. He relies on his daughter to help explain his condition and medical history.

Persona 2: Caretaker (Ms. Tan)

Age: 32

Occupation: Marketing manager

Tech savviness: High

Ms. Tan is a working adult and the main caregiver for her father. She helps manage his medical information across different healthcare providers. She collects paper reports, photos, and PDFs from many sources. She often needs to explain her father's chronic conditions and other health problems to different doctors again and again. This takes

time and increases mental stress. She worries about missing important information and wants a clear way to organize records and track health changes over time.

Scenarios

Persona 1: Patient (Mr. Tan)

[Persona is doing]: Mr. Tan has just received a lab report from his recent hospital visit and feels anxious because he does not understand the results.

[Takes/Opens]: He takes out his smartphone and opens the patient-centered health app.

[Clicks on]: He clicks on the “Upload Document” button to take a photo of his lab report. After uploading, he clicks on “View Summary” where the app shows simplified explanations of his blood sugar and blood pressure readings.

[Persona is doing]: He reviews the color-coded charts that indicate which values are within a healthy range.

[Clicks on]: He clicks the “Share with Caregiver” option to notify his daughter that the report is available for review.

Persona 2: Caretaker (Ms. Tan)

[Persona is doing]: Ms. Tan receives a notification from the app that her father’s lab report has been uploaded. She wants to make sure all his recent medical records are organized.

[Takes/Opens]: She opens the app on her tablet and navigates to her father’s “Health Timeline” view.

[Clicks on]: She clicks on the newly uploaded lab report to view the details and simplified explanations. She then clicks on “Add Notes” to include her observations and reminders from the doctor.

[Persona is doing]: She reviews past reports and compares them with the new results to track trends in her father’s blood sugar and blood pressure.

[Clicks on]: She clicks on “Share with Doctor” to securely send a complete set of recent records before her father’s next consultation.

Best way to perform user research

1. Contextual Inquiry and Observation

What it is: Observing users in their natural environment while they perform tasks related to health management.

Why it works:

For Mr. Tan, you can see how he handles medical documents, uses his phone, and interacts with healthcare staff.

For Ms. Tan, you can understand the stress points of gathering documents, explaining health history to providers, and managing schedules.

How to do it:

Visit patients at home or accompany caregivers during hospital visits.

Take notes on how documents are handled, when confusion arises, and what workarounds they use.

2. Semi-Structured Interviews

What it is: Guided conversations with key questions but leave them flexible to explore unexpected issues.

Why it works:

Captures emotional context (anxiety, frustration) and unmet needs.

Allows patients to express what is hard for them without overwhelming them with unfamiliar jargon.

How to do it:

Ask about daily routines, document handling, interactions with multiple providers, and use of technology.

For Ms. Tan, focus on workflow challenges and the cognitive load of managing her father's health.

3. Diary Studies / Self-Reporting

What it is: Users log their interactions and experiences over a period of time.

Why it works:

Chronic disease management is ongoing; snapshots from one visit may miss critical issues.

Helps track patterns like missed appointments, forgotten measurements, or repeated explanations to doctors.

How to do it:

Mr. Tan can take quick notes or photos of reports at home.

Ms. Tan can keep a short daily log of tasks, challenges, and communication with healthcare providers.

4. Usability Testing with Low-Fidelity Prototypes

What it is: Test app mockups with users performing realistic tasks.

Why it works:

Evaluates whether Mr. Tan can navigate simple functions (uploading photos, viewing summaries).

Tests if Ms. Tan can efficiently organize, annotate, and share multiple records.

How to do it:

Use paper prototypes or simple interactive digital prototypes.

Give task scenarios like “Upload a lab report and share it with your caregiver” or “Compare last week’s readings with today.”

5. Co-Design / Participatory Design Workshops

What it is: Involve users in designing solutions rather than just observing.

Why it works:

Leverages Ms. Tan’s expertise in managing her father’s records and Mr. Tan’s perspective as a patient.

Generates ideas that are realistic for their actual routines and abilities.

How to do it:

Conduct small sessions with paper sketches or post-its to prioritize features (e.g., document aggregation, home monitoring uploads).

Find awareness: Trigger -> Awareness (ads at the clinics/ hospitals etc??) -> Engagement

Possible Pain Points:

> user upload blurry image

>

Elevator pitch

5 Minute pitch